Then, their H.C.F. is x. So, $x = 13$.

$\therefore$ The numbers are $(15 \times 13$ and $11 \times 13)$, i.e., 195 and 143.

Ex. 11. ***Two numbers are in the ratio of 3: 4. Their L.C.M. is 84. Find the numbers.*** **(S.S.C., 2010)**

Sol. Let the numbers be $3x$ and $4x$. Then, their L.C.M. $= 12x$.

So, $12x = 84$ or $x = 7$.

$\therefore$ The numbers are 21 and 28.

Ex. 12. ***The H.C.F. of two numbers is 11 and their L.C.M. is 693. If one of the numbers is 77, find the other.*** **(P.C.S., 2009)**

Sol. Other number $= \left(\frac{11 \times 693}{77}\right) = 99$.

Ex. 13. ***The sum of two numbers is 462 and their highest common factor is 22. What is the minimum number of pairs that satisfy these conditions?*** **(M.A.T., 2004)**

Sol. *Let the required numbers be 22a and 22b.*

Then, $22a + 22b = 462 \Rightarrow a + b = 21$.

Now, co-primes with sum 21 are (1, 20), (2, 19), (4, 17), (5, 16), (8, 13) and (10, 11).

$\therefore$ Required numbers are $(22 \times 1, 22 \times 20)$, $(22 \times 2, 22 \times 19)$,
$(22 \times 4, 22 \times 17)$, $(22 \times 5, 22 \times 16)$, $(22 \times 8, 22 \times 13)$ and $(22 \times 10, 22 \times 11)$.

Clearly, the number of such pairs is 6.

Ex. 14. ***The sum and difference of the L.C.M and H.C.F. of two numbers are 592 and 518 respectively. If the sum of the numbers be 296, find the numbers.*** **(Section Officer's, 2006)**

Sol. Let L and H denote the L.C.M and H.C.F of the two numbers.

Then, $L + H = 592$ ...(*i*) And, $L - H = 518$ ....(*ii*)

Adding (*i*) and (*ii*), we get: $2L = 1110$ or $L = 555$.

$\therefore H = 592 - 555 = 37$.

So, H.C.F. = 37 and L.C.M. = 555.

Let the numbers be x and $(296 - x)$.

Then, $x(296 - x) = 555 \times 37 \Rightarrow x^2 - 296x + 20535 = 0$

$\Rightarrow x^2 - 185x - 111x + 20535 = 0 \Rightarrow x(x - 185) - 111(x - 185) = 0$

$\Rightarrow (x - 185)(x - 111) = 0 \Rightarrow x = 185$ or $x = 111$.

Hence the numbers are 111 and 185.

Ex. 15. ***Find the greatest possible length which can be used to measure exactly the lengths 4 m 95 cm, 9 m and 16 m 65 cm.***

Sol. Required length = H.C.F. of 495 cm, 900 cm and 1665 cm.

$495 = 3^2 \times 5 \times 11$, $900 = 2^2 \times 3^2 \times 5^2$, $1665 = 3^2 \times 5 \times 37$.

$\therefore$ H.C.F. $= 3^2 \times 5 = 45$.

Hence, required length = 45 cm.

Ex. 16. ***Find the greatest number which on dividing 1657 and 2037 leaves remainders 6 and 5 respectively.*** **(Section Officer's, 2006)**

Sol. Required number = H.C.F. of $(1657 - 6)$ and $(2037 - 5)$ = H.C.F. of 1651 and 2032

```
1651 ) 2032 ( 1
       1651
       ----
        381 ) 1651 ( 4
              1524
              ----
               127 ) 381 ( 3
                     381
                     ---
                      ×
```

$\therefore$ Required number = 127.

Ex. 17. ***Find the largest number which divides 62, 132 and 237 to leave the same remainder in each case.***

Sol. Required number = H.C.F. of $(132 - 62)$, $(237 - 132)$ and $(237 - 62)$

= H.C.F. of 70, 105 and 175 = 35.

Ex. 18. ***The H.C.F. of two numbers, each having three digits is 17 and their L.C.M. is 714. Find the sum of the numbers.*** **(C.P.O., 2007)**